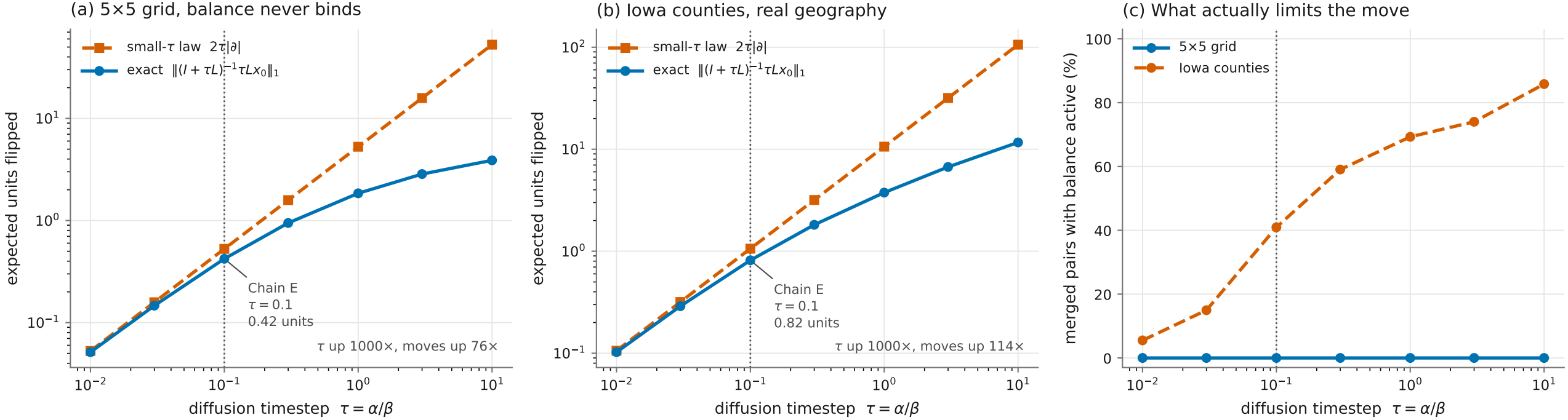
(a) 5×5 grid, balance never binds
small-τ law 2τ|∂|
exact ‖(I + τL)⁻¹τLx₀‖₁
expected units flipped
Chain E
τ = 0.1
0.42 units
τ up 1000×, moves up 76×
diffusion timestep τ = α/β
(b) Iowa counties, real geography
small-τ law 2τ|∂|
exact ‖(I + τL)⁻¹τLx₀‖₁
expected units flipped
Chain E
τ = 0.1
0.82 units
τ up 1000×, moves up 114×
diffusion timestep τ = α/β
(c) What actually limits the move
5×5 grid
Iowa counties
merged pairs with balance active (%)
diffusion timestep τ = α/β